

AI and Job Displacement: A Realistic Plan for Slovenia's Future

Introduction: The Coming AI Job Disruption

Advances in artificial intelligence are poised to reshape labor markets around the world, and Slovenia is no exception. Nearly half of today's jobs globally could be affected by AI in some way, and in advanced economies that figure rises to about 60%^[1]. Roughly half of those impacts may be positive (higher productivity and new tasks), but the other half could see reduced demand for human workers^[1]. This wave of automation promises huge economic efficiency, yet it raises the specter of widespread job displacement, lower wages in some fields, and greater inequality if we do nothing. Slovenia's small, open economy and skilled workforce mean it stands to benefit from AI-driven productivity—if harnessed correctly. However, like many EU countries, Slovenia also has an aging population and existing issues (for example, high housing costs in Ljubljana) that could amplify the pain of rapid job losses. The challenge ahead is ensuring AI's prosperity is shared by all and preventing social upheaval and mass unemployment. History shows that when unemployment climbs even into the low double-digits (10% or more), societies often experience protests, unrest, and surges in crime^[2]. We must therefore be proactive. Many solutions have been proposed to tackle AI-related job loss; this opinion paper will examine the major proposals, explain why they often fall short in general (with examples from larger countries), and consider how those shortcomings might be even more acute for Slovenia. In the end, it argues that the only truly realistic and future-proof response is to embrace **AI-human augmentation** as the core strategy to navigate the AI era.

Solution 1: Universal Basic Income and Cash Guarantees

The idea: Provide everyone with a guaranteed income to buffer against job loss. A popular version is Universal Basic Income (UBI) – for example, entrepreneur Andrew Yang in the US has suggested giving every adult around \$1,000 per month, funded by taxes on big tech profits. A related approach is the **Negative Income Tax (NIT)**, which would top up incomes for the lowest earners through the tax system rather than sending every person the same check. Others include wage subsidies (the state paying part of workers' wages to encourage companies to keep hiring) and wage insurance (temporarily compensating workers for some of the pay lost if they have to take a lower-paying job after being displaced). The logic of all these policies is to maintain people's purchasing power and livelihood even if AI eliminates their jobs – in short, use public funds to ensure no one falls below

a basic income floor. This would, in theory, share the “AI bounty” with everyone and prevent poverty in a high-unemployment future.

Why it fails: These cash-based solutions sound compassionate, but they face severe economic and practical problems. First is the risk of **inflation**. If everyone suddenly has, say, an extra €700 each month unconditionally, demand for goods will increase – but if the supply of essentials (housing, energy, food, healthcare) doesn’t rise equally, prices will simply go up. The extra cash can get eaten by higher rent and grocery bills. In a country like Slovenia, where housing supply (especially in cities) is limited, a blanket UBI could just bid up apartment prices without truly improving living standards. Essentially, you can’t **paper over real scarcity with printed money** – if AI causes real output (jobs, goods) to shrink, just giving out cash doesn’t magically maintain abundance. Small-scale basic income experiments support this concern: for example, Finland ran a trial giving unemployed people €560 a month, and while recipients reported lower stress and better well-being, **there was no improvement in employment rates**^[3]. In other words, people didn’t quit working in a lazy stupor (a common fear) – but UBI also didn’t help them find jobs, nor did it create new jobs for them. Those positive Finnish results came in a controlled, temporary experiment; scaling such a policy up to an entire economy for the long term is a far bigger leap, one that could spur price increases if not carefully funded.

That brings us to the second problem: **funding and sustainability**. UBI or generous income guarantees cost enormous sums. Governments like Slovenia’s would struggle to raise that money without heavy new taxes or debt. Taxing “AI companies” to pay for UBI is often suggested, but as we will see in the next section, that runs into its own obstacles. Absent a magical new revenue source, a UBI means either cutting other services, raising general taxes significantly, or printing money – all difficult choices. A small economy can’t print its own currency freely without risking inflation or devaluation, and Slovenia’s ability to deficit-spend is constrained by EU rules and the euro framework. In practical terms, a broad UBI might force a value-added tax hike or other burdens that themselves hurt consumers. Negative income tax or wage subsidies, which target only lower-income households or certain sectors, might cost less than a full UBI, but they still require substantial government expenditure. Slovenia would essentially be trying to redistribute a pie that, in an AI-induced recession, might be shrinking. This raises the question: *where* will the state get the money to maintain everyone’s income? If the plan is to tax the very AI industries that caused the automation (say, a “robot tax” or a levy on AI companies), those industries might simply leave or find loopholes – a dilemma we explore soon.

Third, **work incentives and fairness** issues arise. If a universal payment guarantees a basic living, **why should people work at all?** This concern may be overstated (many people *want* to work for purpose and extra income, not just survival), but it can’t be ignored. Some opponents ask: if

someone can get, for example, €1,000 per month for doing nothing, wouldn't many choose leisure over low-paid labor? Society could end up with labor shortages in essential but unpleasant jobs, unless those wages rise a lot – which again feeds back into higher costs and taxes. Even with a negative income tax (which phases out as you earn more), there can be “trap” thresholds where taking a higher-paying job leaves you barely better off after the benefit reduction, discouraging advancement. This ties into the **fairness** critique: people contributing more might feel it's unfair that someone who doesn't work receives a similar income. For instance, a working person earning a low wage might see that someone who is unemployed is getting almost the same net income after government top-ups, and resent it. In a small society like Slovenia, such sentiments can be strong – people notice if their neighbor is “getting paid to do nothing”. That resentment can erode political support for these programs. The notion of fairness is also at play with NIT or wage subsidies: *should* profitable companies be indirectly subsidized to employ people, or should the market dictate which jobs survive? Some would argue it's unfair to tax one part of the economy heavily to pay others to essentially stay in obsolete roles or to stay home.

Finally, variants like **wage insurance and portable benefits** suffer from being reactive band-aids rather than true solutions. Wage insurance – say the government temporarily pays 30% of your previous salary if you must take a pay cut in a new job – might soften the blow of displacement, but it relies on the assumption that AI will generate big new profits to fund those payouts. That only works long-term if the AI-driven economy produces enough surplus to support the unemployed or underemployed population. If AI simply replaces workers but doesn't greatly boost overall economic output (at least initially), then wage insurance is drawing from a well that isn't filling. In fact, it has a **reversed logic** problem: the scheme needs the AI to have already *created* extra wealth (or savings) to compensate the person who lost their job to that AI. We end up always a step behind – the person loses their job *first*, then we try to redistribute from whatever gains AI produced, which may not equal the loss. It's a bit like trying to pay for the damage after the fact, rather than preventing it. Portable benefits (making health insurance, pensions, and other benefits tied to the individual rather than their employer) are a good policy in general for a gig-economy world, but they **don't prevent job loss** – they just make it easier to switch jobs. In the face of *no jobs available*, having portable benefits is small comfort. In summary, income guarantee policies can alleviate poverty in theory, but they do not stop the structural problem of job displacement and may introduce new issues (inflation, high taxes, social friction). For Slovenia, a broad UBI or similar would be extremely hard to finance and might simply not deliver the utopian outcome its proponents hope for.

Solution 2: Taxing the AI Winners (Windfall Taxes and Dividends)

The idea: Since the benefits and profits of AI will accrue largely to big tech companies, AI developers, and those who own the capital, one proposal is to heavily **tax the winners** and redistribute the gains to the rest of society. This category includes ideas like an “AI tax” or **windfall profit tax** on AI companies, data dividends (where companies pay you for using your data to train AI), or special levies on robots and automation. Some have suggested a **destination-based tax** system – taxing companies based on where their AI services are used (so they can’t avoid tax by moving their headquarters). Others talk about international coordination, for example an EU-wide tax on AI or digital services, so that companies have no easy escape. The goal here is to capture a share of the enormous wealth that AI is expected to generate and use it to fund social safety nets (like UBI, retraining programs, wage insurance, etc., as discussed). Essentially: *make the tech giants pay for the disruption they cause.*

Why it fails: In practice, taxing highly mobile global companies is notoriously difficult. If a single country tries to impose a hefty new tax on AI revenues or robots, a likely outcome is that businesses will **relocate or find loopholes**. We’ve seen this pattern before – multinational companies routinely shift their operations to wherever taxes and regulations are most favorable. For example, Ireland became a tech hub in part by offering very low corporate tax rates, drawing in companies that wanted to minimize tax. In the Balkans, Serbia has recently attracted international firms by offering a flat 15% corporate tax (one of the lowest in Europe) and other cost advantages[4]. Some companies from higher-tax EU countries have moved parts of their business to Serbia to cut costs by 30–55%[5]. This illustrates a general rule: if Country A gets too expensive or taxing, businesses can shift to Country B. A small country like Slovenia has limited leverage – if it unilaterally slapped a big “AI tax” on companies operating domestically, many would simply choose not to deploy their new AI projects or investments in Slovenia. They might route their activities through other countries or just avoid offering certain AI services to Slovenians to escape the tax. That could leave Slovenia worse off (losing out on new tech entirely, along with jobs and innovation that might have come with those companies).

What about a coalition approach, like the entire EU taxing AI? In theory, a bigger market acting together has more clout. The EU has indeed considered and enacted digital taxes and strict AI regulations (the upcoming EU AI Act, for example). However, even **EU-wide policies can backfire** if companies conclude that compliance is too costly. A recent real-world example: OpenAI’s CEO Sam Altman warned in 2023 that his company might **pull ChatGPT out of Europe** if EU regulations became too stringent[6]. In late 2024, OpenAI launched a new AI product (a text-to-video generator), but

initially **did not make it available in the EU** – likely because they were “still working through compliance” with European rules[7]. In other words, rather than pay for expensive adjustments or risk legal issues, a company can choose to geo-block a service in that region. The upshot for Europeans is that they simply lose access to advanced AI tools. This is the opposite of what we want – it’s like trying to protect workers by banning machinery, but then your workers and firms fall behind foreign competitors who **are** using the new machines. Imagine a Slovenian carpenter forced by local rules to stick with 19th-century hand tools while a carpenter in another country uses state-of-the-art power tools – clearly the latter will outproduce and outsell the former. Over-taxing or over-regulating AI in one region could lead to a similar handicap.

Even if a tax could be implemented, there’s a question of **effectiveness and sufficiency**. Suppose the EU (or a global agreement, if one could dream of that) put a special tax on AI companies’ profits to fund social programs. How do we ensure those companies don’t just pass on the cost to consumers, or cut jobs elsewhere to maintain their profit? If, say, an AI robot tax is charged per robot deployed, a company might respond by simply finding ways to use fewer but more efficient robots, or by cutting human workers to save money for the tax – an unintended negative outcome. And will the tax revenue truly be enough to compensate every displaced worker? The scale of job loss might dwarf the extra revenue, especially if the taxes themselves dampen the growth of AI companies (leading to lower profits to tax). There’s also a **timing issue**: heavy taxation could **squeeze the very profits that are supposed to pay for the safety nets**. In a worst case, you tax the golden goose so much that it stops laying eggs – the AI sector’s growth slows, and there’s less surplus to redistribute, while jobs have already been lost to automation globally.

For Slovenia specifically, **acting alone with taxation is a non-starter**. The country is too small to dictate terms to tech giants or to the flow of capital. If Slovenia tried a unilateral “AI levy,” companies could shift their activities to nearby markets (Italy, Austria, Croatia, etc.) and still reach Slovene consumers via the internet or imports, thereby sidestepping the tax. Slovenia would then collect little revenue but possibly drive away investment. Coordinated EU action is the only plausible way for a tax approach to have teeth – but as discussed, even the EU’s considerable market power has limits when faced with globally mobile tech and the high appetite for AI’s benefits. Moreover, if the EU as a whole became hostile to AI businesses through punitive taxes or regulations, it could end up isolating itself technologically and economically. That would harm countries like Slovenia by depriving them of cutting-edge services and making their industries less competitive internationally. The **bottom line** is that while taxation and redistribution will likely be part of any policy mix (we do need to capture some AI gains for the public good), **taxation alone won’t solve job displacement**. It cannot guarantee full employment or truly stop

companies from automating. At best, it shifts money around; at worst, it drives technology and investment away, leaving people with neither jobs nor advanced tools. No tax scheme can create new human work as fast as AI might eliminate it.

Solution 3: Reskilling, Retraining, and Education

The idea: Whenever technology disrupts jobs, a common response is *“workers can transition into new jobs if they gain the right skills.”* Thus, many experts and policymakers call for major investments in **education and retraining** programs. This could mean re-educating mid-career workers for entirely different professions (an office manager becomes a healthcare technician, for example), expanding vocational training, promoting STEM skills, or encouraging displaced workers to “learn to code” or otherwise acquire skills in fields where labor is still needed. The optimistic vision is that while AI automates some tasks, new kinds of jobs will emerge – so if we upskill the workforce continuously, people can move into those new roles. Proposals here often include government-funded training grants, partnerships with industry for apprenticeships in tech, lifelong learning accounts for each worker, and so on. Essentially: don’t fight the tide of technology, but ride it by teaching people to do the jobs that AI can’t (yet) do.

Why it fails: The harsh reality is that AI in the 2020s is advancing at an **exponential pace**, and retraining humans is comparatively slow and uncertain. The traditional model of retraining assumes a static or at least slowly changing target – for example, a factory worker gets retrained over a couple of years to become an HVAC technician, then works in that field for a decade or more. But what if, by the time our hypothetical worker finishes training and gains some experience, **AI has already started to encroach on that new field as well**? This is not far-fetched: imagine telling truck drivers a few years ago to retrain as warehouse logistics managers – only for AI-driven robots to begin automating warehouses shortly after, or telling an administrative assistant to learn accounting, just as AI starts handling more accounting tasks. Workers could end up in a futile game of leap-frog, always chasing the next “safe” occupation and never catching up. It takes a human years to train and gain expertise in a new domain, whereas AI systems can be upgraded in weeks or months across whole industries. The **time buffer** for retraining is shrinking drastically.

Another issue is **return on investment** – both for the individual and for employers. Older workers, for instance, often find it difficult to embark on a brand-new career in their 50s; even if they do, an employer might be reluctant to hire and train someone who might retire in a decade or be more expensive than a younger worker. Companies themselves have little incentive to retrain employees for new roles if their current role is made redundant by AI. From a purely profit standpoint, if a company can replace a

worker with an AI system or a robot that works 24/7 with lower operating costs, it will likely do so rather than retraining that worker for some other job in the company. The employer might reason: why pay to retrain X as a data analyst (for example) if there are already many analysts in the job market, or if an AI can perform a lot of data analysis faster? Businesses historically prefer to hire already-skilled workers for new needs, or just eliminate roles, rather than invest heavily in retraining their entire workforce for speculative future tasks. This is especially true for smaller companies that can't afford large training programs. As a result, the burden often falls on governments and individuals to retrain – and government programs have a mixed track record. Many retraining initiatives see low completion rates or people completing courses that don't actually lead to new employment because local demand for that skill was overestimated.

Crucially, **AI itself is beginning to challenge the notion of “AI-proof jobs.”** It used to be said that jobs requiring creativity or human empathy – say, an artist or a nurse – would be safe for longer. But AI's capabilities are expanding into creative domains (writing, art generation) and even care roles through robotics. People often assume a “human touch” will always be needed, but experiments suggest humans can form attachments or respond socially to machines more than we expect. For instance, studies have shown that if you give a robot a few human-like attributes – say, a pair of eyes on a screen, a human name, and a backstory – people begin to treat the robot as if it has a personality, even feeling empathy towards it^{[8][9]}. In one experiment, simply adding eyes to a humanoid robot significantly increased how much agency and emotion people attributed to it, which in turn boosts their willingness to accept and cooperate with the robot^{[8][9]}. This hints that even caregiving or customer service roles, which we imagine must be done by people because “only a person can connect with another person,” might be handled by sufficiently socialized AI entities in the future. We already see early examples: therapeutic robot pets for the elderly, chatbot “friends” for lonely individuals, etc. As these technologies improve, the space of totally safe human-only jobs shrinks further.

We should also consider the **breadth of AI's reach**. Much of the recent focus has been on white-collar and knowledge work (AI writing code, drafting documents, analyzing data). Indeed, many of the first impacts of AI have been on office jobs and professional services. Slovenia's workforce includes many such roles (IT, finance, administration) given it's a developed economy – those are directly in the crosshairs of AI software like large language models. But blue-collar and manual jobs are not immune either; they've just had a bit more lag because physical automation is harder. That lag may only be on the order of a decade. AI “agents” are rapidly being integrated with **robotics** – for example, AI vision and coordination enabling robots to do basic warehouse picking, or self-driving vehicle technology threatening driving jobs. An assembly-line worker or a warehouse picker might think switching to a construction trade or truck driving is a safer bet than an office

job, but **self-driving trucks and robot bricklayers** are actively being developed. Some experts estimate that jobs involving routine physical tasks have maybe a 5-15 year buffer before advanced robotics catches up. That's not a very long time. Telling someone to retrain from, say, a manufacturing job to a healthcare job might work for a while – until AI and robotics start making serious inroads into healthcare support roles and diagnostics as well.

In Slovenia, a further limitation is the **scale and availability of new high-tech jobs**. It's one thing to retrain people if there is a booming new industry ready to absorb them. But if AI is not only taking away jobs but also making certain new ventures extremely efficient with few workers (think of a factory largely run by robots and a few engineers), the number of replacement jobs might be far smaller than the jobs lost. Slovenia can try to cultivate new industries (like green energy, AI development, etc.), but realistically the numbers might not match up. Without massive new job creation, retraining simply shuffles workers around while total employment drops. In addition, not everyone can or wants to be a software developer or a data scientist – and even those roles are being assisted or partly replaced by AI (developers using AI code assistants require fewer colleagues to accomplish the same work, for example).

In summary, **traditional retraining is too slow and too uncertain to keep up with AI**. It's not a bad idea – continuous learning is part of any healthy economy – but as a primary solution to potential mass unemployment, it is woefully inadequate. We cannot count on moving the entire workforce up the skill ladder continually, when the top of the ladder itself is being lifted by AI. Slovenia should indeed improve education and help workers adapt, but if we rely solely on “educate people for the next job,” we may find there isn't a “next job” (at least not a human-only one) by the time we finish the schooling. The real retraining needed is not just learning a different human skill – it's learning to work alongside AI (we'll return to that in the final section). Humans must become *adaptive* learners in an AI world, but expecting them to outrun the pace of machine advancement is unrealistic.

Solution 4: Work-Time Reduction and Job Sharing

The idea: If AI allows the same output with fewer workers, one straightforward response is to **share the remaining work** among more people by reducing individual work hours. Rather than having, say, one person working 40 hours and another person unemployed, you could have both work 20 hours and each do half the job. This concept has been floated in forms like a **four-day workweek**, a six-hour workday, or more flexible part-time arrangements. Unions and some policymakers historically have supported work-time reductions as a way to reduce unemployment – indeed, France famously implemented a 35-hour full-time work week in the early 2000s with an aim (in part) to spur hiring. The intuition is that if productivity

per worker rises (which AI would do for those who remain employed), workers could keep the same pay for fewer hours, or accept proportionally less pay for fewer hours, thereby freeing up “time slots” to employ others. Some also argue shorter workweeks improve quality of life and could make people more resilient to job loss (by having more free time to train or search for other gigs).

Why it fails: In practice, mandating or widely adopting shorter hours as a solution to job loss runs into economic and logistical snags. If companies are required to pay the **same salary for fewer hours** (to avoid reducing workers’ income), this is effectively a sharp increase in labor cost per hour. Businesses faced with that might react by either raising prices (if they have market power), which can hurt consumers, or by trying to offload work to machines even faster (since the relative cost of a human just went up). If instead workers accept **proportionally lower pay for fewer hours** to make room for others, then each worker has less income – which could mean they need to find a second part-time job to make ends meet. But if jobs are generally scarce, that second job might not exist, or they’re competing with another person who also needs a second job. We could end up in a situation where many people are underemployed and still struggling, rather than a few being unemployed – arguably a fairer distribution of hardship, but still hardship.

There is also the issue of **total work vs. total demand**. If AI causes a 20% drop in needed human labor to produce everything society needs, theoretically you could cut everyone’s working hours by 20% and employ the same number of people as before. However, if AI causes a 50% drop, or 80% drop, it becomes very hard to spread the remaining work thin enough. People can’t all work 1 hour a day; at some point, the overhead of having many employees doing tiny bits of work is impractical for businesses. Companies might prefer to keep a smaller workforce full-time for efficiency and just let others go. And if **aggregate demand** (consumption) falls because many people have lost income, there might not even be enough work to distribute. For example, if AI leads to goods being super cheap or to companies not needing to produce as much (since tasks are optimized), simply slicing the work pie differently doesn’t help if the pie is shrinking.

We can look at real examples: France’s 35-hour workweek did not eliminate unemployment – companies found ways to adapt, sometimes by increasing work intensity or cutting overtime, and unemployment remained an issue. In many cases, firms didn’t hire a bunch of extra workers; they simply asked employees to get the same amount of work done in 35 hours (through higher efficiency or pressure) as they did in 40, or they adjusted shifts without significantly expanding staff. It turns out businesses won’t hire extra people unless there’s a clear benefit to having more bodies; just freeing up hours doesn’t guarantee a new hire if one person can still cover the essential tasks. Likewise, proposals for 4-day weeks often assume productivity will remain

about the same (so workers get 100% pay for, say, 80% time). If that holds, fine – but then where is the “extra” job creation? If productivity doesn’t hold and output falls, then either the company suffers or it needs to hire more staff (which is the goal for employment). But hiring more staff to do the same work means higher labor cost, which not all businesses can afford unless perhaps the government provides a subsidy for those extra hires.

Speaking of subsidies, one could imagine the state stepping in: perhaps the government could compensate workers for the lost day of pay in a 4-day week (so they maintain a full salary for 80% work) – effectively a state-sponsored job-sharing scheme. However, that becomes very expensive (it’s almost like a partial basic income) and again relies on tax revenue from a robust economy, which might be exactly what’s in jeopardy during an AI transition. It circles back to the funding problem of Solution 1.

From the **worker’s perspective**, reduced hours are lovely if income is preserved, but not if income drops. Many Slovenians, like people elsewhere, have fixed costs (rent or mortgage, bills, etc.) that won’t proportionally decline just because their hours did. If their pay is cut 20%, they may not be able to cover their expenses unless those also somehow drop – which they likely won’t, especially if AI isn’t bringing cost of living down uniformly (consider housing costs, which tend not to drop without major policy interventions). So, workers might end up **needing two jobs**: one might imagine a person working 4 hours at Company A and 4 hours at Company B to make up a full day’s worth of pay. But if there were two half-jobs available per person, why wouldn’t Company A or B just consolidate that into one job? The scenario only helps if every employer specifically only wants part-timers. In reality, many roles benefit from continuity and experience that come with full-time positions.

In Slovenia’s context, mandating or encouraging significantly shorter workweeks could make Slovenian firms less competitive internationally unless everyone else is doing the same. A factory in Slovenia that runs on two 6-hour shifts with a full extra set of workers might have higher unit labor costs than a factory in, say, Slovakia or China that runs a single 12-hour shift with automation and fewer workers. Unless productivity soars to compensate, simply dividing work among more people can increase costs and reduce a business’s ability to invest or even survive against global peers. Slovenia cannot isolate itself from competitive pressures – if work-sharing significantly raised costs or lowered output, businesses could outsource or automate more to cope.

In conclusion, **work-sharing and shorter hours are not a panacea for technological unemployment**. In theory, they spread work more evenly, which is socially appealing. In practice, without careful coordination and likely subsidies, they tend to either fizzle (with no big uptick in hiring) or create new problems like underemployment and hidden financial stress. At

best, reduced working time might improve quality of life for those with jobs – a worthy goal in itself – but it doesn’t guarantee jobs for those who would otherwise be laid off. In an AI-driven labor market crunch, telling everyone to “just work less and share jobs” is an oversimplification. It fails to address situations where there simply isn’t enough *useful* work for humans to do, even if we cut hours to the bone. Thus, while work-time reduction might be part of a broader conversation about the future of work (perhaps as a means to distribute productivity gains), it **isn’t a solution to job displacement** on its own.

Solution 5: Steering or Stopping AI Development

The idea: Rather than adjusting society to AI, adjust AI to society. This perspective includes attempts to **steer AI development** toward complementing humans instead of replacing them, and even calls to *pause or ban* certain types of AI that are seen as too dangerous for jobs. Influential voices like researcher Erik Brynjolfsson have suggested we focus on “human-in-the-loop” AI that works alongside people, and change incentives so companies build tools to empower workers rather than autonomous replacements. On the more extreme end, some have floated global moratoriums on advanced AI research (not unlike nuclear non-proliferation treaties) to prevent AI from ever reaching a stage where it can outdo humans broadly. In policy terms, steering AI could involve regulations that, for example, prohibit fully automated services in certain sectors or require that a human worker must be retained in the process. It could also mean heavy-handed measures like outright banning advanced autonomous systems (an analogy being how some countries ban dangerous weapons or certain biotech). The aim here is to **protect jobs by limiting AI’s scope** or by guiding innovation so it augments rather than automates.

Why it fails: This approach faces a fundamental issue of global coordination and economic incentive. Stopping or significantly slowing AI development would require *all the major players* (countries, big tech companies, research institutions) to cooperate and agree on limits. We have a precedent in nuclear weapons control, but even that is only partially successful and came about because nuclear war is universally catastrophic with no economic upside. AI is different: it offers enormous economic and strategic benefits *to those who lead in it*. It’s more analogous to an industrial revolution than to a nuclear bomb. In a competitive world, if Country A decides to halt AI progress, Country B will eagerly continue, knowing they can leap ahead economically and technologically. There’s a **prisoner’s dilemma** or “race” dynamic – everyone might secretly wish for a slow down to avoid disruption, but no one wants to be the only one to slow down and thus fall behind. The United States, China, the EU, and others are all investing heavily in AI precisely because they don’t want to be left out of what is seen as a new foundational technology. Even companies: if a few top companies agreed to

self-regulate and not push certain boundaries, some start-up or a foreign competitor may not play along and will seize the advantage.

The comparison to nuclear arms is instructive: nuclear weapons were ultimately seen as pure threats (their use is destructive to all sides, which inspired treaties). AI, on the other hand, is already delivering improvements – in medicine, science, productivity – which are not things people want to ban globally. There is no clear line between “dangerous AI that takes jobs” and “useful AI that saves lives or makes money.” For instance, an AI that can fully drive a truck and put drivers out of work is the same AI that can make trucking safer and faster. Would we ban self-driving technology to save driving jobs? If one country bans it, another will develop it and sell cheaper transport services or cars that outperform the others. Unlike with nuclear bombs, using advanced AI benefits the user economically, so the incentive to cheat on any global ban is huge. Even within Europe, we see tension: the EU’s forthcoming AI Act tries to steer AI (by banning certain high-risk uses and enforcing transparency), but companies like OpenAI have balked, and some fear it will just mean Europe gets AI later or in a reduced form compared to the US or Asia[6].

For Slovenia specifically, **its influence on AI development is minimal**. Slovenia is not a leading AI creator; it mostly will be an adopter of AI developed elsewhere. If Slovenia alone decided, “We will not allow advanced AI within our borders,” it would isolate itself from technology that other nations and businesses are using. That could make Slovenian industries non-competitive. Imagine Slovenia banned automated customer service chatbots to save call center jobs – Slovene companies would then either have to pay more for large call center staff (raising their costs relative to foreign competitors using bots), or they’d outsource the work abroad where bots are legal, achieving nothing for local employment. Similarly, if Slovenia said “no autonomous trucks on our roads,” but neighboring countries allow them, eventually logistics routes might just go around or through Slovenia with minimal local involvement, and Slovenian transport companies would lose business.

The idea of **steering AI toward complementarity** rather than substitution is noble. Research and funding can be directed to projects that enhance human work. For example, investing in AI that helps doctors diagnose rather than replaces them, or AI that assists teachers in the classroom instead of automating teaching. This is somewhat feasible at a policy level – grants and incentives can encourage certain tech development. However, it’s difficult to truly separate “complementary” from “substitutive” AI. Many technologies can be used in either way. A robot could work in a factory alongside a human (augmenting them by doing heavy lifting) or it could be upgraded to eventually handle the human’s tasks too. The market ultimately drives what happens: if a fully autonomous system becomes cheaper and more effective, companies will adopt it, regardless of original intentions. Unless there is a

strict law forbidding fully autonomous operation, companies will push AI as far as possible to save costs. And if there *is* a strict law (say, “banks must maintain X% of staff and not fully automate customer service”), firms might comply domestically but still find ways to cut labor via outsourcing or using AI in areas not covered by the law. It can also stifle innovation – a start-up in Slovenia might avoid developing a cutting-edge AI solution if the environment is hostile to automation, and that start-up might simply move to a more permissive jurisdiction.

In short, **development will not stop worldwide** as long as AI offers competitive advantages. Global bans are implausible because the upsides are too high for any one defector. Coordination at the EU or international level to only pursue “friendly AI” may produce some guidelines and ethical frameworks (which is good) but won’t entirely prevent the creation of job-killing AI elsewhere. And Slovenia by itself has virtually no say in this global race – it must adapt to the technologies that bigger players create. Waiting and hoping that someone “steps on the brakes” universally is wishful thinking. The more likely scenario is AI continues to improve rapidly, and any attempt to wall off a country from it (to protect jobs) would do more harm than good to that country. Therefore, relying on halting or heavily steering AI as the main strategy to save jobs is not realistic. We can and should advocate for responsible AI (for safety, privacy, etc.), but we can’t count on “saving jobs by decree” in a world where so many forces are pushing the other way. It’s like trying to hold back a tide with a fork – the scale is just too great. We need solutions that assume *AI is coming full force* and figure out how society can thrive in that reality.

Solution 6: Strengthen Worker Protections and Unions

The idea: If market forces won’t protect workers, then stronger **labor protections, unions, and legal frameworks** might. This solution involves updating labor laws to fit the AI era: perhaps requiring companies to negotiate with unions before implementing AI that could cause layoffs, mandating advance notice or compensation for workers replaced by automation, or establishing new rights like “the right to retrain” or severance funded by automation gains. Unions could push for contract language that limits the use of AI in certain jobs or guarantees a minimum number of human workers. Governments might also bolster social safety nets – like more robust unemployment benefits or job transition assistance – which, while not preventing job loss, at least protect workers from destitution. The overarching idea is **collective bargaining and regulation** can slow down the pace of displacement and ensure workers aren’t simply tossed aside. Some also argue for mechanisms like employee representation in decisions about tech adoption (for instance, a works council that must agree to major technological changes in a factory).

Why it fails: The main problem here is **speed and rigidity**. Laws and unions tend to move gradually, negotiating incremental changes and implementing policies after long deliberation. AI's growth, on the other hand, is blisteringly fast and accelerating. By the time a new worker protection law is drafted, debated, passed, and implemented, the labor market might look completely different. We've seen analogous situations with gig economy jobs – regulations to protect gig workers (like rideshare drivers) lagged years behind the spread of those platforms, and even now are patchy. With AI, the lag could be even more severe because the transformation might occur over just a few years. **Waiting for laws to catch up is a losing game** when AI is improving on a monthly timescale. There is also the political angle: enacting strong labor protections often faces pushback from business interests and can be watered down. If countries try to enforce very strict rules (like “no layoffs if company is profitable” or similar), companies often respond by finding loopholes (relabeling employees as contractors, relocating operations, automating in ways that aren't covered by the letter of the law).

Collective bargaining can certainly help workers get better terms – for example, a union might negotiate retraining support or a delay on automation. But unions are not omnipotent, especially in the private sector where global competition is a factor. If a union in Slovenia makes it too hard to implement a certain AI, a multinational might simply shift that work to a subsidiary in another country with less resistance. Or the company may argue that without automation, it cannot survive against competitors, and thus convince even the workers that it's automation or bankruptcy (this has happened in various industries historically, where unions had to concede to technology changes to keep companies alive). Furthermore, **bargaining structures are often too rigid** to adapt to rapid changes. Unions typically negotiate contracts every few years; AI's capabilities could evolve multiple generations within a single contract period. Unless agreements are extremely flexible and updated constantly (which is not how contracts usually work), they'll always be addressing the last wave of tech, not the next one.

Another angle is that strong worker protections (like making layoffs very difficult or expensive) can have **unintended side effects**. While they might slow down firing, they can also slow down hiring – companies become cautious in adding staff if they know it's hard to remove them. In an AI context, a firm might choose not to hire a replacement for someone who leaves, but rather to automate that function, because hiring a new human comes with a commitment and potential legal hurdles to later adjust. Or a start-up might avoid hiring many employees in Slovenia if it perceives labor laws as too constraining, preferring to base its workforce in a more flexible environment. It's a delicate balance: too much rigidity can stifle business agility and thus economic growth, which ultimately hurts workers too.

For Slovenia, which already has relatively strong worker rights compared to many countries, adding even more layers (like requiring government or union approval for deploying certain AI systems) could deter foreign investment. A small country has to be careful – if it's seen as overly restrictive, businesses (especially in tech) might not set up shop there. That would mean fewer jobs *to begin with*, ironically. Moreover, enforcement is key: laws on paper don't automatically protect workers if companies can circumvent them. For instance, if a law said "you must keep X number of employees on payroll even after automating," a company might comply nominally but slowly shift those people to trivial roles or encourage them to leave voluntarily (through attrition or offering buyouts).

It's important to note that **worker protections are still valuable** – they can ensure fairness, prevent abuse, and provide some cushion. But as a **core strategy to prevent job displacement**, they are insufficient. They act as a seatbelt at best, not an avoidance of the crash. If AI is set to eliminate a large chunk of jobs, strong labor laws might delay the impact for a short time or soften the blow for some, but they won't ultimately save those jobs unless alternative roles are created. And if pushed too far (like trying to legally forbid automation outright), they could isolate an economy or cause businesses to fail, which again results in job loss. Collective action through unions can win important concessions – say, shorter workweeks with preserved pay (tying back to solution 4) or retraining funds – but they cannot hold back the tide of technology indefinitely without broader changes in the economy.

In summary, strengthening labor protections should be seen as a **complementary measure**, not a solution by itself. It's part of ensuring a just transition, making sure workers have some power and safety in the process, and that the gains from AI aren't solely captured by company owners. However, if a nation relies on these protections as the main defense against technological unemployment, it will likely find itself perpetually behind the curve, adjusting rules after the damage is done. In the worst case, overly rigid rules could even hasten job loss by discouraging employers from keeping roles human when they can automate and avoid labor law complications. As the saying goes, we can't legislate away reality – and the reality is that AI's impact on jobs will be huge and fast, while laws are slow and often reactive.

(Transition: After reviewing six major approaches and their shortcomings, it's clear that none provides a silver bullet. Each idea – from UBI to retraining to regulation – has merits and should play some role in a multifaceted strategy. But each on its own is riddled with holes and constraints, especially in a small country context. We likely need a different mindset altogether to tackle the AI job shock. Rather than fighting against the current or trying to patch the

symptoms, what if we turn AI into an advantage for every worker? This leads us to the final - and arguably only - robust solution.)

Final Solution: AI-Human Augmentation as the Core Strategy

After eliminating the usual suspects, we arrive at the one strategy that stands a real chance against the pace and scale of AI disruption: **AI-human augmentation**. In plain terms, this means proactively equipping people with AI tools and skills so that each worker becomes far more productive and can do things even advanced AI alone cannot. Instead of AI replacing the human, the human **uses** AI to perform their job better. This approach turns the narrative on its head – rather than humans being made redundant by machines, humans *upgrade* themselves by integrating machines into their work. It's a vision where every professional, from factory workers to doctors to teachers, has an "AI assistant" or co-pilot that amplifies their abilities.

Why it's the realistic solution: Augmentation is essentially the **only strategy that scales with AI's exponential growth** because it harnesses that very growth. If you can't beat them, join them. By giving workers access to AI, you ensure that as AI gets smarter, so do your workers (via the tools they use). It's a way to keep human labor relevant and valuable. Crucially, this is already happening organically in some fields – most notably in software development. Many programmers today use AI coding assistants (like GitHub Copilot or similar tools) to write and debug code. The results are striking: developers using these AI helpers can complete tasks significantly faster – one study found by **about 55% faster on average** for a common coding task^[10]. In essence, a coder with AI is much more productive than a coder without. Companies have noticed: firms are actually encouraging their engineers to use these tools because it boosts output and efficiency^[11]. This is telling – unlike previous solutions which businesses often resist (no company *wants* higher taxes or strict firing laws or being forced to pay UBI), augmentation is something **employers and workers both can embrace**. Employers get higher productivity; workers get to keep a role (albeit a transformed one) and often find their job more engaging (let the AI handle the drudge work, while the human tackles more creative or complex parts).

Think about other professions: a doctor with an AI diagnostic system can see more patients or catch conditions she might have missed, effectively **multiplying her capability**. A teacher with AI-assisted tutoring programs can personalize learning for each student in ways one teacher alone never could. A marketing specialist with AI can draft campaigns and analyze customer data in a fraction of the time, freeing them to focus on strategy and client interaction. Even in trades: a mechanic with an AI diagnostic tool can pinpoint issues much faster; a farmer with AI-driven drones and sensors can manage more acreage with the same labor. In each case, the human is still in the loop, but their output is amplified by the machine. The key is that

the human role shifts – often towards more supervision, creative judgment, interpersonal interaction, or complex decision-making, with AI doing the repetitive or highly data-intensive tasks.

This approach is also **relatively cheap and quick to implement** compared to massive social programs. Consider the cost comparison: providing an individual with access to a powerful AI tool (like a subscription to a state-of-the-art AI service) might cost on the order of €10–€30 a month. Training them to use it effectively might be a short course or just learning by doing, since many modern AI tools are designed to be user-friendly (you can chat with them, etc.). By contrast, supporting that same individual with a basic income could cost €700+ a month, or a multi-year college retraining program could cost tens of thousands of euros. Augmentation is **incredibly cost-effective**. For what a government might spend on one month of UBI for a person, it could potentially cover a year of that person's AI tool subscription. It's essentially leveraging the economies of scale of software: one AI service can serve many users at low marginal cost.

Moreover, augmentation is **future-proof in a way retraining isn't**. We don't know exactly which jobs will exist in 5 or 10 years, but we do know that whatever they are, they will likely involve working with AI systems (because AI will be pervasive). By training people *now* to be adept at using AI in their workflow, we make them adaptable to any new job environment that comes. It's a form of meta-skilling: learning how to quickly learn and collaborate with AI. For instance, a person who today learns how to use AI to analyze data and generate reports can tomorrow apply that skill in a different domain because the core skill is interacting with AI effectively. This beats trying to guess "which specific technical skill should I learn that won't be automated?" Instead of betting on a specific safe job, bet on **human+AI as the paradigm for all jobs**. It's a way of not putting all eggs in one occupational basket.

From the perspective of **society and social stability**, augmentation has huge advantages. If most people remain employed in productive work – even if the nature of work changes – you avoid the scenario of large-scale unemployment that could lead to unrest. People retain a sense of purpose and agency. We sidestep, or at least postpone, the hard question of "what if there are no jobs for people?" by ensuring that people always have a competitive edge: themselves *plus* the AI. In effect, each worker becomes, say, five times as productive with AI – which might mean a company needs fewer total workers, *unless* they can expand output because now their augmented workforce can do so much more (which is likely in a growing economy; more productivity can lead to lower prices or new markets, which create more demand for output and hence can sustain employment). Even if some job reduction happens, it will be far less dramatic than if AI just outright replaces humans without augmentation. And those humans who do

lose a particular job can more easily find another if *everyone* is expected to use AI tools anyway – their experience collaborating with AI is transferrable.

Slovenia, specifically, is well positioned to adopt an augmentation strategy. It has a well-educated workforce with strong technical literacy (by global standards), decent digital infrastructure, and it's small enough that a national program to boost AI usage could be rolled out fairly swiftly. For example, the government or private sector could ensure that every student and worker has access to AI assistance: imagine if libraries, schools, and job centers provided free or subsidized access to the latest AI systems and training on how to use them. The country could include "AI literacy" in curricula and adult education. Employers in Slovenia could be encouraged (or even given tax incentives) to provide AI tools to their employees and train them to integrate these into their tasks. This way, Slovenian companies become **more competitive**, not less – they'd be on the cutting edge of productivity. Rather than trying to shield old jobs, Slovenia would be creating the **augmented jobs of the future**, essentially jobs that might not exist elsewhere yet. In doing so, it could attract investment instead of repelling it: firms might set up operations in Slovenia precisely because the workforce is AI-savvy and can achieve high output. It's a chance to turn a potential weakness (being a small economy that can't dictate global trends) into a strength (being nimble and forward-thinking, adapting faster than larger countries).

To those who might worry that augmentation is just a slower path to the same problem (eventually AI might not even need the human at all), consider time scales and the value of *buying time*. Even if, say, in 30-40 years AI could do literally everything a human+AI team can do today (which is speculative), giving a whole generation of workers meaningful employment in the interim is crucial. Society needs time to adjust, maybe to develop new economic models if full automation ever becomes feasible. Augmentation provides a **preventative approach** – it addresses the problem now, rather than curing its symptoms later. It's much easier to keep someone employed by evolving their job as tech evolves, than to let them fall into unemployment and then try to pull them out later with welfare or retraining. In medical terms, it's prevention versus cure: far more effective and cheaper to vaccinate (augment workers) than to treat a widespread illness (mass unemployment) after it spreads.

One more benefit: augmentation naturally leads to sharing the gains of AI. If every worker is using AI, the productivity gains are distributed across the whole workforce, not concentrated in a few AI tech companies. Yes, the AI providers still profit, but workers and their firms capture a lot of value too (through higher output and potentially higher wages if their skills are in demand). It's a way to avoid a sharp inequality jump between those who own AI and those who don't – because everyone, in a sense, would "own" an AI helper or at least have access to one. Imagine each person essentially

having a tireless personal assistant or research analyst at their side. That's a far more optimistic future than one where 90% of people are idle on stipends and 10% (plus robots) do all the work.

In conclusion of this section, AI-human augmentation is not just a theoretical win-win – we're already seeing its early signs in real life, and it's the **only strategy that aligns incentives**. Workers can support it (it makes their jobs easier, not eliminated), companies support it (it improves profits and productivity), and governments should support it (it averts mass unemployment and keeps society stable). It takes the very force that threatens jobs and uses it to empower those jobs. Of course, making this the cornerstone of policy requires action: widespread training, ensuring access to AI tools (maybe subsidies or public options for expensive software), and a mindset shift in education and workplaces. But these are achievable tasks, especially compared to trying to fund an entire population's income indefinitely or getting dozens of countries to agree on halting AI research. Augmentation is actionable right now: for the cost of one month of basic income for a person, you could provide them with AI assistance for years^[12]. It's hard to overstate how game-changing that cost-benefit ratio is. Simply put, **to save jobs, we must make AI a tool in everyone's hand, not an enemy at the gates**.

Conclusion: Empower People Now, Don't Wait for the World to Change

The landscape of solutions to AI-induced job displacement is often dominated by grand schemes and hopeful theories – but as we've seen, each popular solution has critical flaws when examined closely. Unconditional basic incomes risk inflation and are hard to fund; taxes on AI are easily dodged and can backfire; retraining alone can't outrun the AI treadmill; work-sharing only shuffles scarcity around; trying to halt AI is unrealistic globally; and stronger labor rules, while helpful for fairness, won't halt the tide and can be too slow. These ideas aren't *wrong* in principle – many of them should be pursued in moderation (for example, a modest safety net, sensible regulation, some retraining efforts, etc., can all help). But even in combination, at their very best they are partial and **suboptimal**, buying a bit of time or easing some pain but not truly solving the core problem. Relying on them as the primary plan would be like patching holes in a dam that's about to be overwhelmed by a flood.

The core problem is that AI's growth is **exponential**, while our societal responses (political processes, education systems, international agreements) move linearly or even glacially. We simply **won't catch up with laws or treaties** if we try a business-as-usual approach^[13]. Waiting for some global epiphany – whether it's all countries agreeing to stop AI development, or a switch from capitalism to a different system – is dangerously naive given the speed of change. By the time such a revolution in governance happened (if it

ever did), we could have a decade or more of upheaval and unemployment behind us, with all the human cost that entails. History tells us that even a sustained unemployment rate above 10% can tear at a society's fabric^[2]. We're talking protests in the streets, loss of faith in institutions, maybe worse. And that threshold could be crossed in the not-too-distant future if AI adoption outpaces job creation. Even a theoretical utopia where nobody *needs* to work and everyone can pursue leisure – a vision some paint for an AI-rich future – would require a difficult transition period. You don't go from a work-based society to a leisure-based one overnight without turmoil. People need purpose, income, and stability in the interim decades, not just an assurance that "someday it'll be fine".

Therefore, the **bottom line** is clear: we must focus on **practical, immediate measures** that empower individuals to thrive alongside AI, rather than hoping for broad structural change to bail us out. AI-human augmentation is exactly that kind of measure. It doesn't require global coordination or massive ideological shifts – it's something every country, and indeed every worker, can start doing now. It treats AI as an ally to be embraced, not merely a threat to be regulated or a profit source to be skimmed. By giving "the small man" – the average person – a fighting chance with AI at their side^[13], we make the playing field more even. We democratize the power of AI so it's not just the tech elites or giant corporations that have it. This strategy is **preventative**: it aims to prevent the worst outcomes (mass unemployment, extreme inequality) by getting ahead of the curve. It's much like vaccination in public health – a little effort and cost upfront can avert a catastrophe later.

Slovenia, and any country in a similar position, should not sit back and wait for the dust to settle. The dust will not settle by itself – we have to actively shape the outcome. Augmentation may not solve every single challenge of the AI revolution, but it squarely addresses the central issue: keeping people meaningfully employed and productive. It buys us time and stability to work out longer-term adaptations (like how education, economy, and social contracts might evolve in an AI-heavy world) without the society-breaking shock of sudden mass joblessness.

In conclusion, the path forward is not about one magic policy or resisting change; it's about **adapting at the speed of technology**. Equip people with AI, train them to use it well, and cultivate an economy where human creativity and judgment paired with machine efficiency and intelligence is the norm. Such a workforce can continuously find new value to add, even as AI takes over old tasks. This way, AI becomes a tool for human prosperity rather than a harbinger of human obsolescence. We owe it to ourselves and the next generation to choose this proactive route. The challenges are immense, but so is human ingenuity – especially when amplified by our own creations. Instead of fearing the rise of intelligent machines, let's focus on **raising the capability of our people**. The sooner we do that, the better

we can ensure that the story of AI and jobs in Slovenia (and beyond) is not one of despair, but of empowerment and shared progress.

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